

Development of a Novel Cable-Driven Ankle Rehabilitation Device

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Abstract

As one of the most frequently injured joints in the human body, the ankle is highly susceptible to sprains and fractures that often necessitate assisted movement to restore functional mobility. Although physiotherapists traditionally conduct rehabilitation through individualized, hands-on sessions, numerous robotic rehabilitation systems have been developed in recent years to help in ankle rehabilitation. Despite these advances, the development of ankle rehabilitation devices still faces several open challenges. Thus, this paper introduces the design and the performance analysis of a novel cable-driven ankle rehabilitation device. This device aims to facilitate and accelerate the ankle rehabilitation process. The mathematical model of the proposed device is presented, followed by performance analysis using Unity 3D software. The prototype is described along with its control system and a custom-developed serious game. The feasibility of the novel device is verified through numerical and experimental results over the ankle motion range.

1. Introduction

The ankle joint is a critical joint of the human body. It enables bipedal locomotion, allowing individuals to stand and walk while supporting the entire body weight. However, the ankle joint is also one of the most frequently injured joints with injuries ranging from muscular and ligament damage to bone fractures, often resulting from excessive physical activity [1]. Also, ankle impairments are common in neurological disorders, particularly after stroke, where muscle weakness, spasticity, and impaired motor control significantly compromise joint stability and mobility [2]. These deficits often result in reduced dorsiflexion, abnormal gait patterns such as foot drop, and diminished balance, increasing the risk of falls and limiting functional independence [3]. These injuries can severely impair mobility, compromise daily activities. Ankle rehabilitation is an essential part for restoring the ankle mobility after such injuries. Traditionally, this rehabilitation process is supervised by therapists and specialists by performing protocols/training exercises [4]. This is however time-consuming for both patients and specialists. A specialist can also only supervise a patient at a time. Due to a large number of patients and the lack of specialists, the sessions are therefore short and infrequent, limiting the

effectiveness of the rehabilitation [5]. Robotic rehabilitation devices offer a promising alternative to complement traditional rehabilitation methods [6].

Several robotic rehabilitation device concepts have been developed to assist in ankle rehabilitation. This can both shorten the recovery time of the patient and reduce the workload of the specialists [7]. Table 1 provides a brief comprehensive review of existing ankle rehabilitation robots. The first column lists the corresponding citations, followed by the main technical features: degree of freedom (DOF), supported ankle movements, actuation type, exercise modes, control strategy, ankle axis alignment capability, and implemented safety mechanisms.

The Degree Of Freedom (DOF) refers to the number of independent parameters required to define the position or configuration of the device. Some systems incorporate translational motions in addition to rotations, which explains why certain designs exceed three DOF. With a focus on ankle rehabilitation, the third column of Table 1 specifies the groups of ankle motions and their corresponding Ranges Of Motion (ROM) that each device can generate during therapeutic activities. The actuation type indicates the mechanism used to drive the robot and produce the required movements for rehabilitation. In this context, exercise mode is commonly defined by the presence or absence of muscular effort from the patient. Passive exercises involve no active muscle contraction, with foot/ankle motion fully driven by the device or a therapist. In contrast, active exercises require the patient to voluntarily move the foot/ankle, typically against resistance, to promote muscle strengthening.

The control strategy describes the variables and control approaches selected by the authors to achieve accurate robotic motion and deliver effective rehabilitation. To ensure safe operation and prevent secondary injuries, it is essential that the robot's axis of rotation aligns, at least approximately, with the anatomical ankle axis. Some studies explicitly incorporate this alignment into their design methodology. Some articles present only theoretical models and analyses, whereas others report experimental validation using physical prototypes. To enhance safety and minimize risks arising from control failures, overloads, human errors, or unforeseen events, some systems include dedicated safety mechanisms that can quickly disengage or release the patient's limb. Any information missing or unspecified in the reviewed articles has been left blank in Table 1.

Table 1: Comparison of main features of different ankle rehabilitation robots selected in the literature.

Paper	DOF	Motion and ROM	Actuationtype	Exercise mode	Control strategy	Ankle axis matching capacity	Safety system
[8]	3	DO/PL: 25°/35° IN/EV: 20°/17° AD/AB: ±25°	3 servomotors	-	-	No	No
[9]	4	DO/PL : 30°/45° IN/EV : ± 30° AD/AB : 25°/20°	-	-	-	Yes	-
[10]	3	DO/PL IN/EV : ± 10° AD/AB : ± 10°	4 Linear electric actuators	Active Passive	Position control	-	-
[11]	3	All	2 Linear electric actuators 1 DC Servo	Active Passive	Trajectory tracking control	Yes	-

			motor				
[12]	2	DO/PL IN/EV	2 Stepper motors	Passive	Trajectory tracking control	-	-
[13]	3	All	3 DC Servo motor	Passive	Position- speed control	Yes	-
[14]	1	DO/PL 25°/55°	1 Brushless DC motor	Active Passive Resistive	Position Control	-	Yes
[15]	2	DO/PL IN/EV	2 Linear electric actuators	-	-	Yes	-
[16]	3	All	2 Linear electric actuators 1 DC servomotor	Passive Active Resistive	Trajectory tracking control	Yes	Yes
[17]	1	DO/PL: 20°/45°	Motor	-	Trajectory tracking control	-	-
[18]	4	All	-	Static Dynamic	-	Yes	-
[19]	3	All	Servo Motors	-	-	-	-
[20]	3	DO/PL : 30°/45° IN/EV : $\pm 10^\circ$ AD/AB : $\pm 36^\circ$	2 Linear electric actuators 1 Servo motor	Active Passive	Position control Impedance control	-	Yes
[21]	6	All	6 Linear electric actuators	Active Passive	-	-	-
[22]	1	DO/PL	2 PMA*	-	-	-	-
[23]	3	All	-	-	-	-	-
[24]	3	DO/PL : 30°/46.2° IN/EV : $\pm 23.5^\circ$ AD/AB : $\pm 36^\circ$	2 Linear electric actuators 1 Stepp er motor	Active Passive	Impedance control	Yes	-
[25]	2	DO/PL : $\pm 30^\circ$ IN/EV : $\pm 30^\circ$	2 Motor s	-	-	-	-

*Pneumatic Artificial Muscles (PMA)

dorsiflexion/plantarflexion (DO/PL) ankle motion

Inversion/Eversion (IN/EV) ankle motion

Adduction/Abduction (AD/AB) ankle motion

Table 1 presents a concise review of ankle rehabilitation robots reported in the literature from 2019 to 2025. The objective was to examine the design characteristics of these devices and categorize their fundamental functionalities and key technical features. A set of main attributes was identified and used to classify the surveyed systems, Table 1.

Regarding the types of movements supported by each robot, the brief review shows that most devices provide dorsiflexion/plantarflexion (DO/PL) motion. This prevalence is expected, as DO/PL plays a central role in daily activities, especially in gait [24]. Therefore, it is reasonable that most rehabilitation platforms include at least this movement group. However, other motions are also relevant in clinical practice and should be incorporated to enhance therapeutic versatility and better support healthcare professionals [26].

The analysis also revealed a general lack of attention to safety mechanisms. As indicated in Table 1, few reviewed studies explicitly describe safety systems in their designs. A similar trend is observed regarding ankle joint axis alignment that studies clearly address the importance of matching the device's rotational axis to the patient's anatomical ankle axis. This gap highlights a need for greater emphasis on ergonomics and safety in design developments.

The development of ankle rehabilitation devices still faces several open challenges. Ensuring precise alignment between the device's mechanical axis and the patient's anatomical ankle joint remains a critical issue, as misalignment can compromise comfort and therapeutic effectiveness. Most systems continue to offer only limited ranges of motion, highlighting the need for compact, low-cost mechanisms capable of replicating natural multi degree of freedom ankle biomechanics. Additionally, real-time adaptive and personalized therapy is hindered by insufficient integration of intelligent control strategies. Safety mechanisms also remain underdeveloped, particularly in managing excessive torque, abnormal movements, and sudden resistance. Usability challenges persist as well, including the need for improved wearability, portability, and ergonomic comfort during extended use. From a clinical perspective, the lack of standardized evaluation protocols and homecare use limits widespread adoption, while difficulties integrating devices into routine clinical workflows further constrain their applicability. Moreover, patient engagement tools, such as serious games, often lack depth, personalization, and biomechanical relevance, reducing long-term adherence.

Among these devices, the CABLEAnkle [7][19] is a parallel cable-driven lightweight wearable robot designed for ankle rehabilitation. Based on this concept, a novel Cable-Driven Ankle Rehabilitation device was developed. The proposed device permits the patient's anatomical ankle axis, and the use of cable turns the device more safety. Unlike many existing designs, this one permits an additional degree of freedom, the ability to vary the platform height, to a more flexible setup of the device by the health professional, improving the patients comfort. This manuscript presents the kinematic analysis of the novel proposed cable-driven mechanism, followed by the CAD/CAE numerical simulation to evaluate the kinematic model. The prototype and control system are described and validated. For the ankle rehabilitation to be more interesting and entertaining, the cable-driven ankle rehabilitation device is combined with a custom-developed serious game.

2. Materials and Methods

2.1. Anatomy of the ankle joint

The ankle joint is a complex that connects the shank to the foot. This joint, who is a combination of muscles, bones and ligaments, enables the three rotational movements of the foot, Fig. 1. The inversion-eversion is defined as the rotation around the X-axis for the roll movement. The plantarflexion-dorsiflexion is defined as the rotation around the Y-axis for the pitch movement. Finally, the abduction-adduction is defined as the rotation around the Z-axis for the yaw movement [26].

Although these ranges of motion vary among individuals, the overage ankle motion range limits [26] are reported int the Table 2.

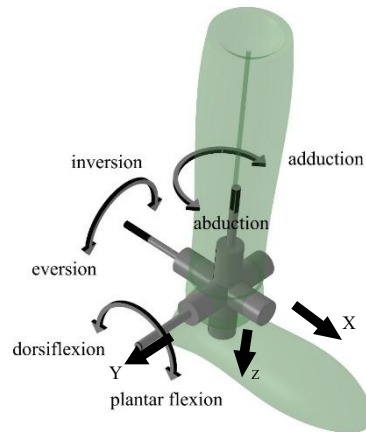


Figure 1 Scheme of ankle movements.

Table 2 Range motion of human ankle [26].

Axis	motion	range
X-axis	Eversion	12°
	Inversion	12°
Y-axis	Plantarflexion	50°
	Dorsiflexion	20°
Z-axis	Abduction	15°
	Adduction	15°

2.2. Design of the Cable-Driven Ankle Rehabilitation device

The proposed Cable-Driven Ankle Rehabilitation device consists of a fixed rectangular cuboid structure and a moving platform, Figs. 2 and 3. The platform is actuated by four servomotor-pulley-cable assemblies, each equipped with a load sensor. In use, the foot is fixed on the moving platform.

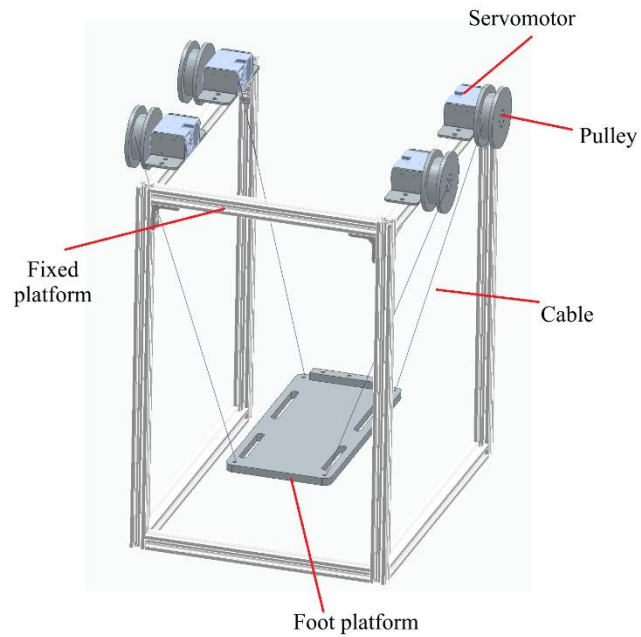


Figure 2 CAD of the structure and the platform.



Figure 3 CAD of a patient using the cable-driven ankle rehabilitation device.

The design consists of two platforms connected by four variable length cables, Fig. 2. The lower platform is the moving platform where the foot is placed while the upper platform is the plan where the fixed position servomotors are connected to the structure, Fig. 4. By adjusting the cable lengths, the servomotors enable the platform to move with four degrees of freedom. The following hypothesis are assumed for the design: the cables are always in tension and don't have axial deformation, the points between the cables and the platforms are considered as spherical joints.

The platform movement is controlled by four Robotis Dynamixel XM540-W270-T servomotors, which are managed via a C# script using the DynamixelSDK library. The interfaces used are a Robotis U2D2 and a Robotis U2D2 Power Hub Board.

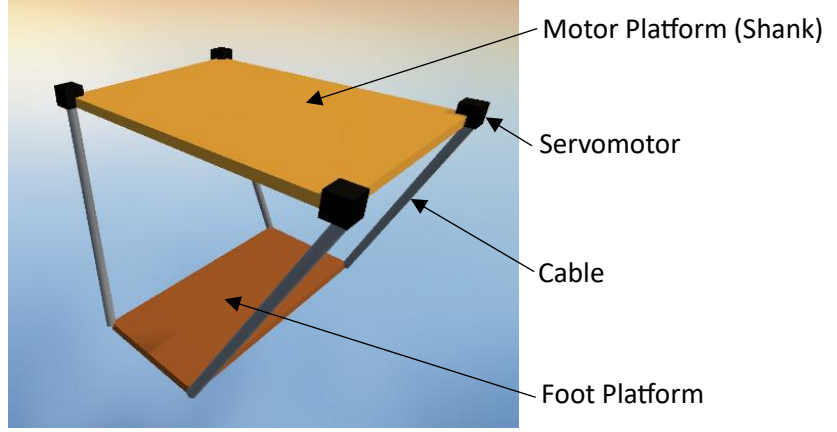


Figure 4 Proposed 3D model of the design using Unity 3D.

2.3. Kinematic analysis of the device

To simplify the mathematical model of the device, the fixed platform was considered in movement according to the moving platform. The newly fixed platform (foot platform) F is represented by the reference frame $Oxyz$ and the newly moving platform (motor platform) M by the reference frame $Ouvw$, O representing the ankle joint, Fig. 5. The point O is assumed to be at coordinates $(0; 0; 0)$. The cable lengths are named l_1, l_2, l_3 and l_4 . The cables upper extremity are attached to the points M_i of the motor platform with i the i^{th} cable, whereas the lower extremity is attached to the point F_i of the foot platform. The M_i and F_i joints between the platforms and the cables are assumed to be spherical joints. The orientation of M is a ZYX Euler Angle rotation and can be expressed by the equation $R = R_z(\varphi)R_y(\theta)R_x(\psi)$. The translation is represented by the variable h .

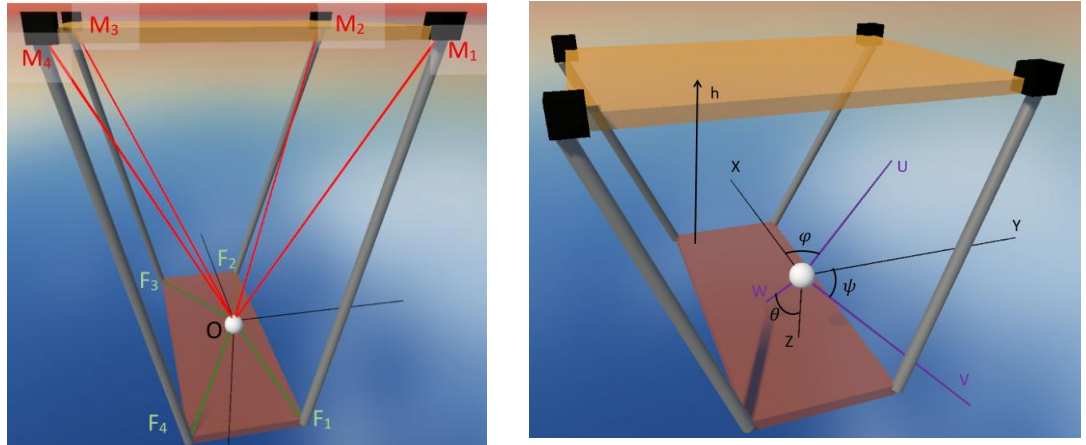


Figure 5 Mathematical model of the design.

The close loop equation of each cable can be expressed by:

$$OM_i + M_iF_i + F_iO = 0 \quad (1)$$

For the proposed device, the geometrical characteristics of the foot platform in $[mm]$ are:

$$F_1 = \begin{bmatrix} -35 \\ 60 \\ 80 \end{bmatrix}, F_2 = \begin{bmatrix} 245 \\ 60 \\ 80 \end{bmatrix}, F_3 = \begin{bmatrix} 245 \\ -60 \\ 80 \end{bmatrix}, F_4 = \begin{bmatrix} -35 \\ -60 \\ 80 \end{bmatrix} \quad (2)$$

So:

$$F = \begin{bmatrix} -35 & 60 & 80 \\ 245 & 60 & 80 \\ 245 & -60 & 80 \\ -35 & -60 & 80 \end{bmatrix} \quad (3)$$

Using the same method, the coordinates of the points of M at height 0 mm and null angles are obtained:

$$h = \psi = \theta = \varphi = 0; \quad M = \begin{bmatrix} -28 & 150 & -285 \\ 195 & 150 & -285 \\ 195 & -150 & -285 \\ -28 & -150 & -285 \end{bmatrix} \quad (4)$$

The Euler rotational transformation matrix of M relative to O can be expressed as:

$$R = R_z(\varphi)R_y(\theta)R_x(\psi) = \begin{bmatrix} \cos(\varphi) \cos(\theta) & -\sin(\varphi) \cos(\psi) + \cos(\varphi) \sin(\theta) \sin(\psi) & \sin(\psi) \sin(\varphi) + \cos(\psi) \sin(\theta) \cos(\varphi) \\ \cos(\theta) \sin(\varphi) & \cos(\psi) \cos(\varphi) + \sin(\psi) \sin(\theta) \sin(\varphi) & -\sin(\psi) \cos(\varphi) + \cos(\psi) \sin(\theta) \sin(\varphi) \\ -\sin(\theta) & \sin(\psi) \cos(\theta) & \cos(\psi) \cos(\theta) \end{bmatrix} \quad (5)$$

The first step to calculate the mathematical model is to transform the rotation of the F :

$$F' = F \times R \quad (6)$$

The M' frame is obtained using the height h :

$$M' = M + \begin{bmatrix} 0 & 0 & h \\ 0 & 0 & h \\ 0 & 0 & h \\ 0 & 0 & h \end{bmatrix} \quad (7)$$

The cable vectors can then be found using the equation:

$$L = F' - M' \quad (8)$$

Finally, the cable lengths l_i are obtained by getting the norm of each cable vector:

$$l_i = \sqrt{(L_{ix})^2 + (L_{iy})^2 + (L_{iz})^2}; i \in [1; 4] \quad \text{knowing that} \quad L = \begin{bmatrix} L_{1x} & L_{1y} & L_{1z} \\ L_{2x} & L_{2y} & L_{2z} \\ L_{3x} & L_{3y} & L_{3z} \\ L_{4x} & L_{4y} & L_{4z} \end{bmatrix} \quad (9)$$

3. CAD/CAE Results and Performance Analysis

Tests were conducted using Unity 3D and PhysX, the 3D Unity's physics engine. The data obtained with the Unity simulation are saved in csv files and plotted using Excel. The convention used for the cable name in the simulations is represented in **Figure 6**:

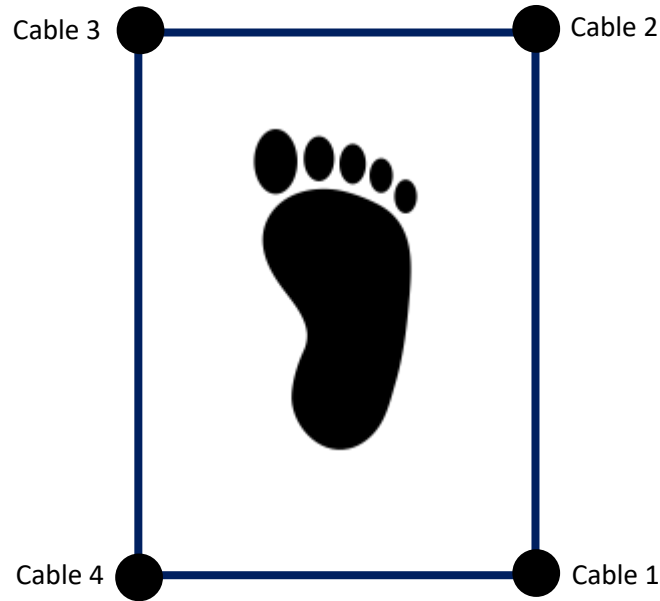


Figure 6 Cable name convention.

The data have been recorded using the Unity's transform position, rotation and scale of the different cables and platforms. The transform position represents the position of the different elements in the world referential, using the ankle as center, the transform rotation is the rotation of the elements, and the transform scale represents the size of the elements, including the length of the cable.

3.1. Height

This section analyzes the variable h , representing the height of the foot platform relative to the ground, Fig. 7. The base height of the foot platform used is the ground with $h=0$. The height is not actively used for the rehabilitation, but is set to a fixed value for the entire rehabilitation session. The height is set to a comfortable value for the patient.

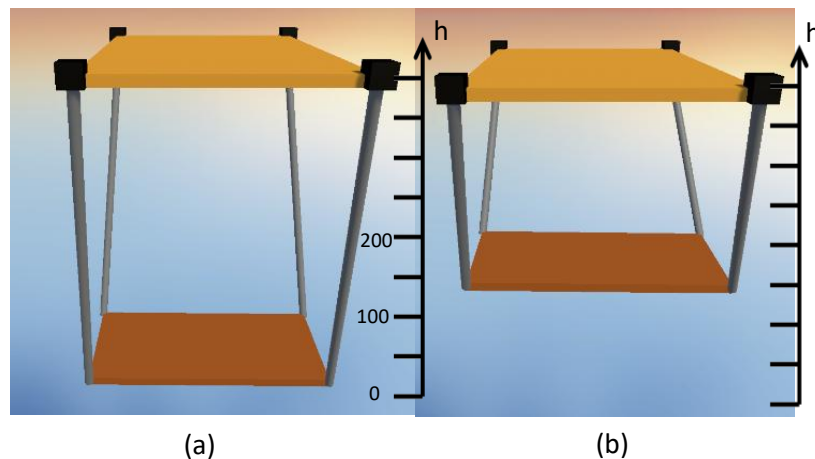


Figure 7 3D model in different height: (a) $h=0$ mm; (b) $h=150$ mm.

For the following records, only the variable h is varying and the three angles are set to 0° , Fig. 8. The height h is also limited to $[0; 300]$ mm due to the geometry of the structure and ankle range movements, Table 2.

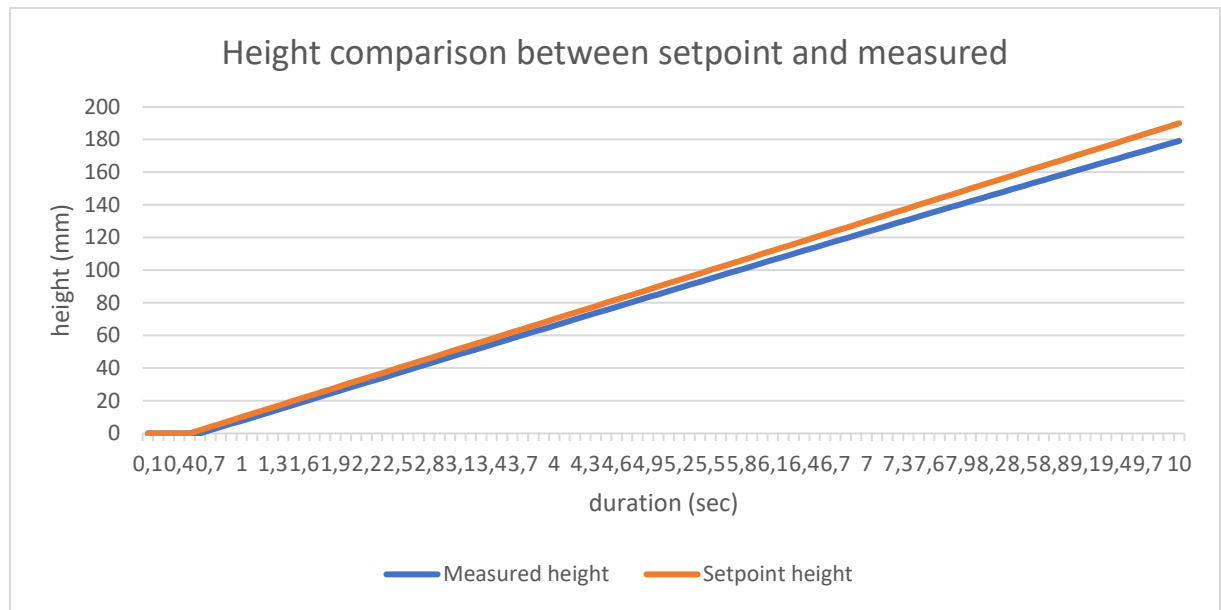


Figure 8 Simulated height comparison between height setpoint and height measured.

It can be noticed that the measured height in the Unity simulation has a delay of around 0.2 seconds to follow the given setpoint. As expected, the length of each cable is approximately the same whatever the height. The length of the cables is also decreasing as the height h is increasing, as expected, Fig. 9.

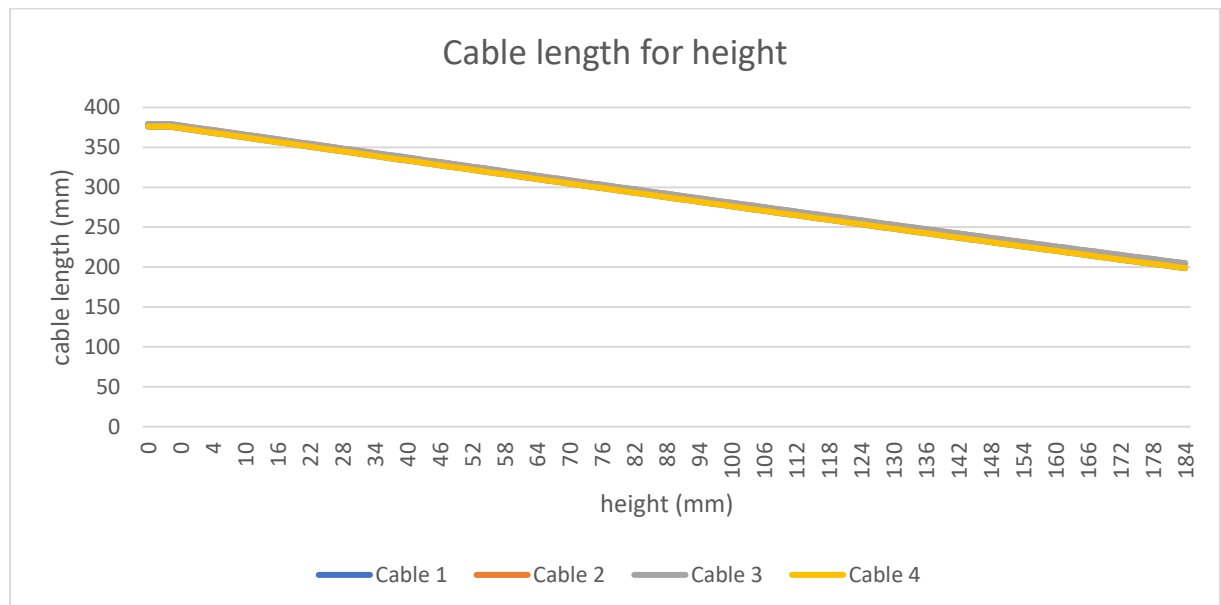


Figure 9 Simulated cable length for height.

3.2. Pitch

The variable pitch corresponding to the dorsiflexion and the plantarflexion movements around the ankle, **Figure 1**. The initial rotation value is set to 0° and is limited by $[-20; 50]^\circ$ in use due to the ankle anatomical limits. For the simulation and to test the mathematical model, an angle range of $[-40; 40]^\circ$ was utilized, Fig. 10.

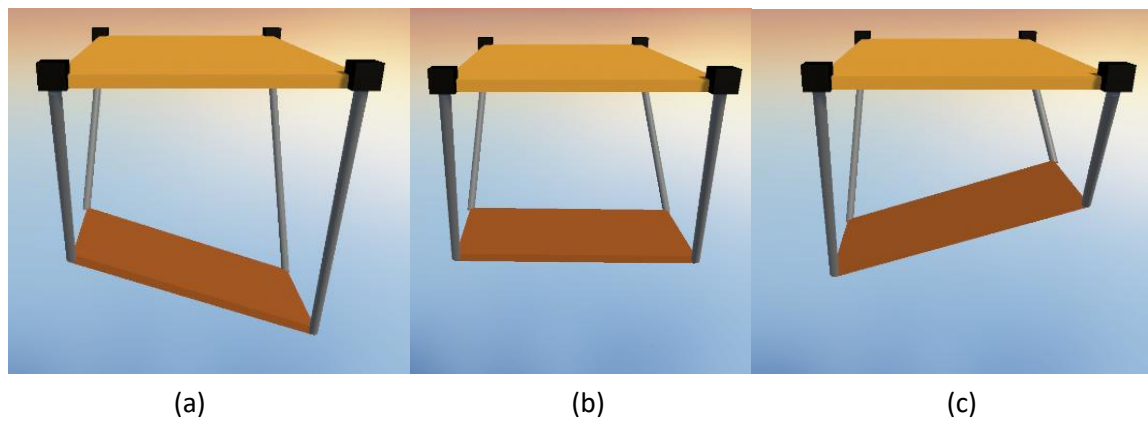


Figure 10 3D model for plantarflexion/dorsiflexion ankle motion: (a) pitch = 20°; (b) pitch = 0°; (c) pitch = -20°.

For the following records, Fig. 11, only the variable pitch is varying while the other angles are set to 0° and the height to 150 mm. The oscillations in the measured angle from the simulation are due to the physics engine of Unity 3D, as the foot platform does not stabilize between two setpoints. It can be noticed that the 1st and the 4th cable have the exact same length whatever is the pitch angle. The 2nd and the 3rd are the same too. As the ankle is not centered in the foot, the cables length of the front of the platform has a more important amplitude, around 250 mm, than the back cables, less than 50 mm. It is coherent as the ankle is located at the back of the foot. The front cables length is decreasing for increasing pitch angles while the back cables are increasing, meaning that the front of the foot is going higher, Fig. 12.

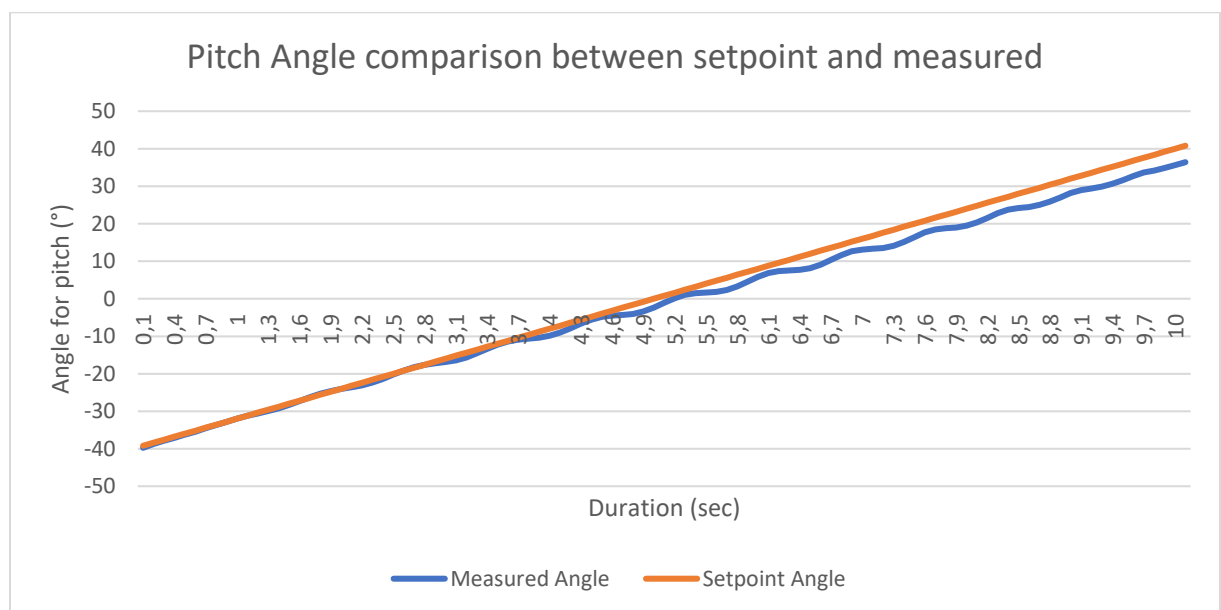


Figure 11 Simulated pitch comparison between pitch setpoint and angle measured.

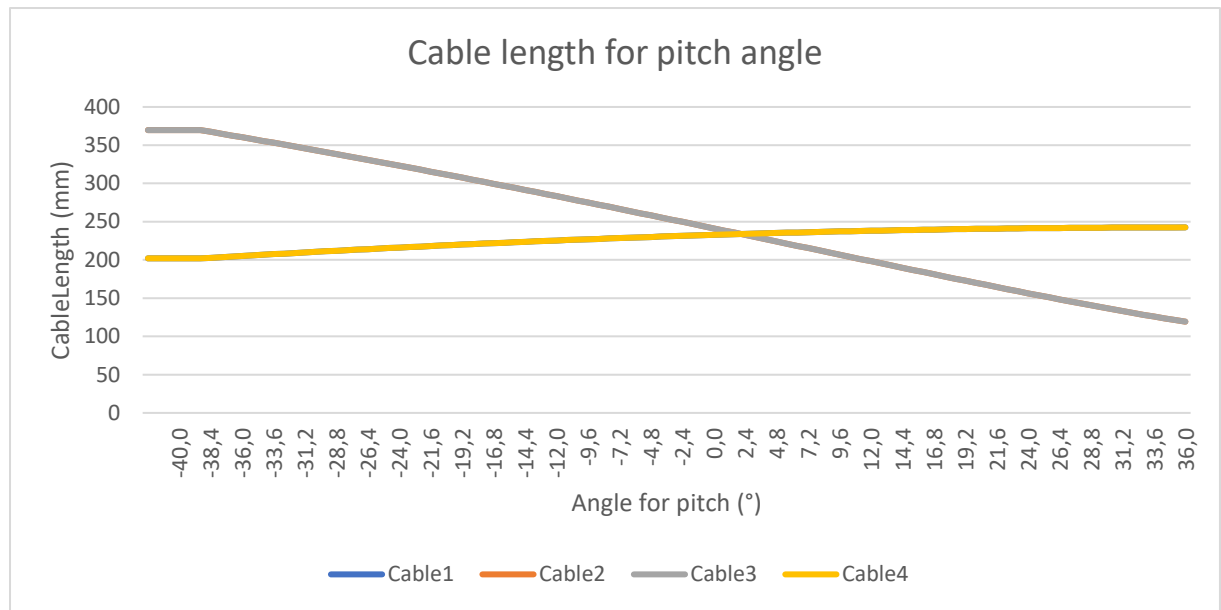


Figure 12 Simulated cable length for pitch angle.

3.3. Roll

The variable roll corresponding to the eversion and the inversion movements around the ankle, **Figure 1**. The initial rotation value is set to 0° and is limited by $[-12; 12]^\circ$ in use due to the ankle anatomical limits. For the simulation and to test the mathematical model, Fig. 13, an angle range of $[-40; 40]^\circ$ was utilized.

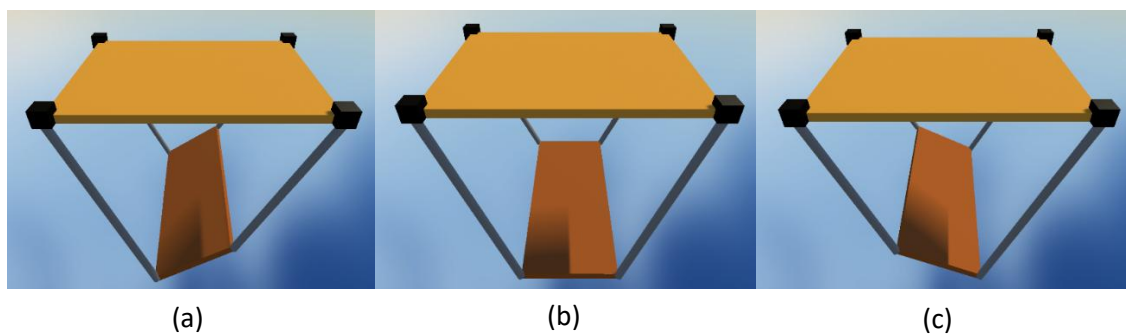


Figure 13 3D model for eversion/inversion ankle motion: (a) roll = 20° ; (b) roll = 0° ; (c) roll = -20° .

For the following records, only the roll variable is varying while the other angles are set to 0° and the height to 150 mm, Fig. 14. It can be noticed that the 1st and the 2nd cable have approximately the same length whatever the pitch angle is. The 3rd and the 4th ones are similar too. For the roll rotation, the ankle is centered in the foot, this can be noticed by the symmetry of the right and left cables according to the sign of the roll angle, Fig. 15.

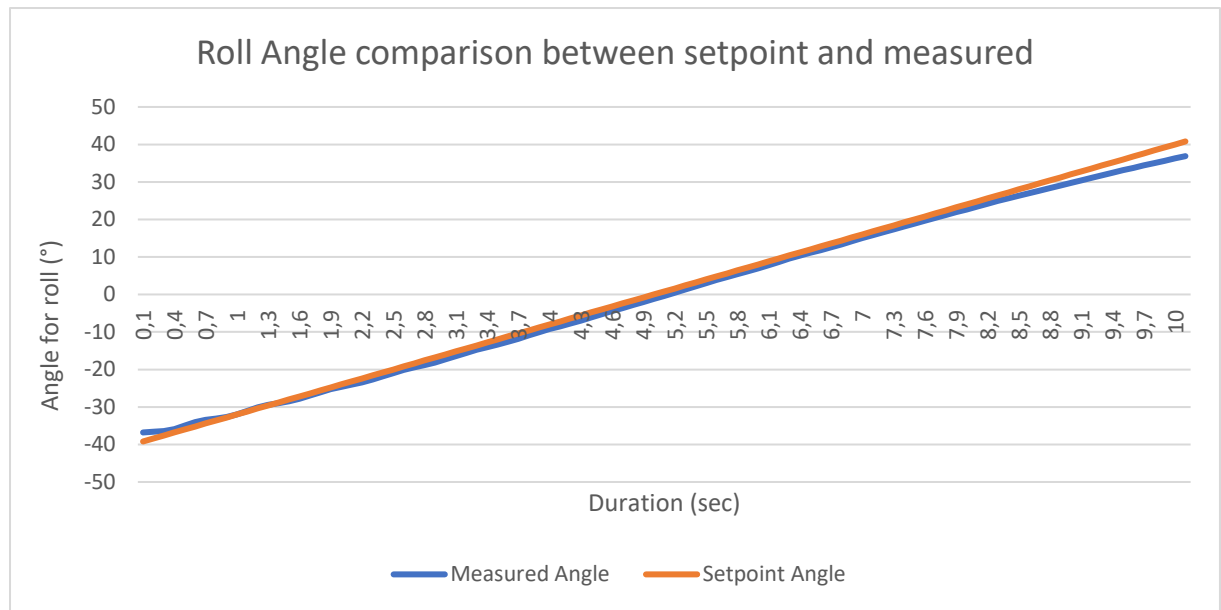


Figure 14 Simulated roll comparison between roll setpoint and angle measured.

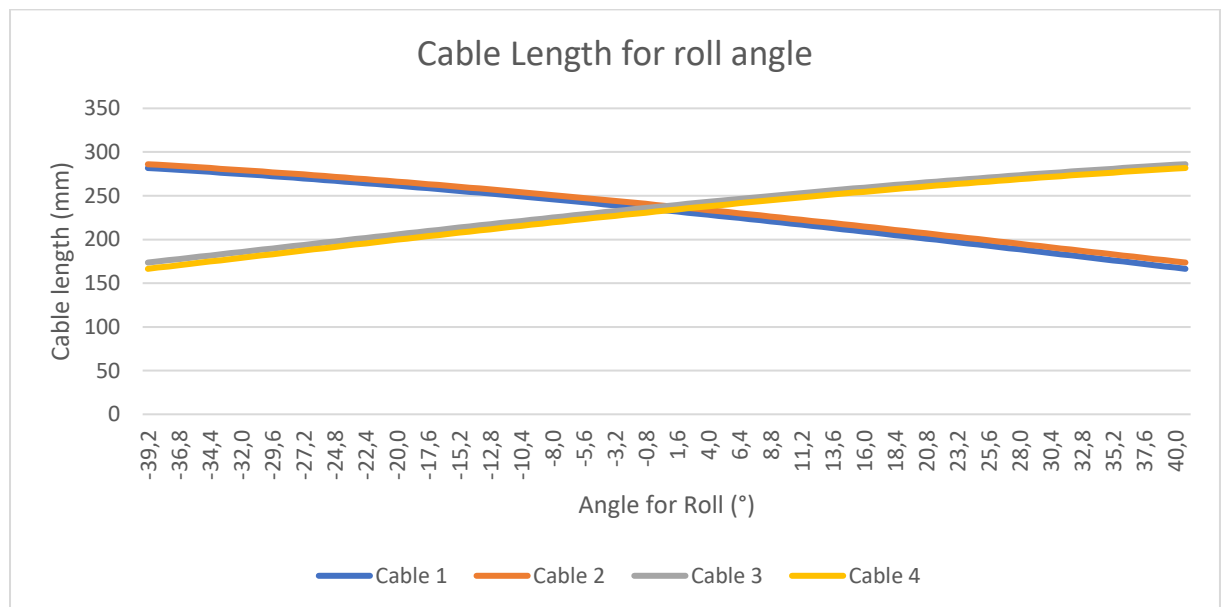


Figure 15 Simulated cable length for roll angle.

3.4. Yaw

The variable yaw, Fig. 16, corresponding to the abduction and the adduction movements around the ankle, **Figure 1** Scheme of ankle movements. The initial rotation value is set to 0° and is limited by [-15; 15]° in use due to the ankle anatomical limits. For the simulation and to test the mathematical model, an angle range of [-40; 40]° was utilized.

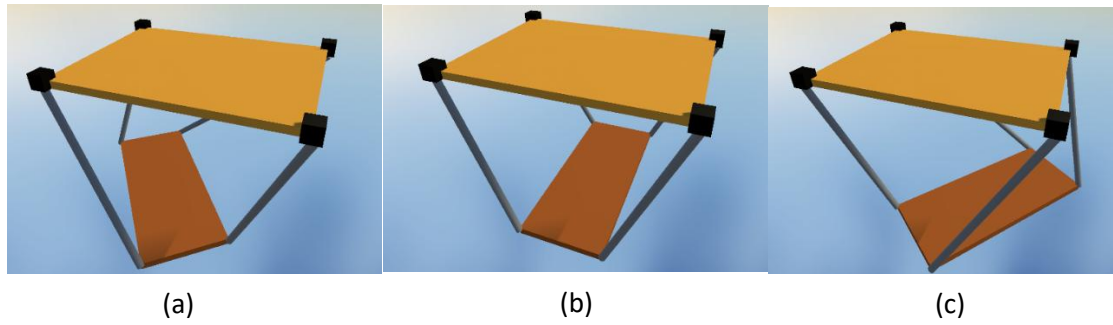


Figure 16 3D model for abduction/adduction ankle motion: (a) yaw = 20°; (b) yaw = 0°; (c) yaw = -20°.

For the following records, only the yaw variable is varying while the other angles are set to 0° and the height again is constant with the value of 150 mm, Fig. 17. It can be noticed that the 1st and the 4th cables are symmetric according to the sign of the yaw angle. In the same way, the 2nd and 3rd cables are symmetric. The two front cables have a more important amplitude than the back cables. The distance between the ankle and the front of the foot is higher than the distance between the ankle and the back of the foot, resulting in a higher rotation radius for the front cables. For an angle of 0°, all the cables have the same length, as expected, Fig. 18.

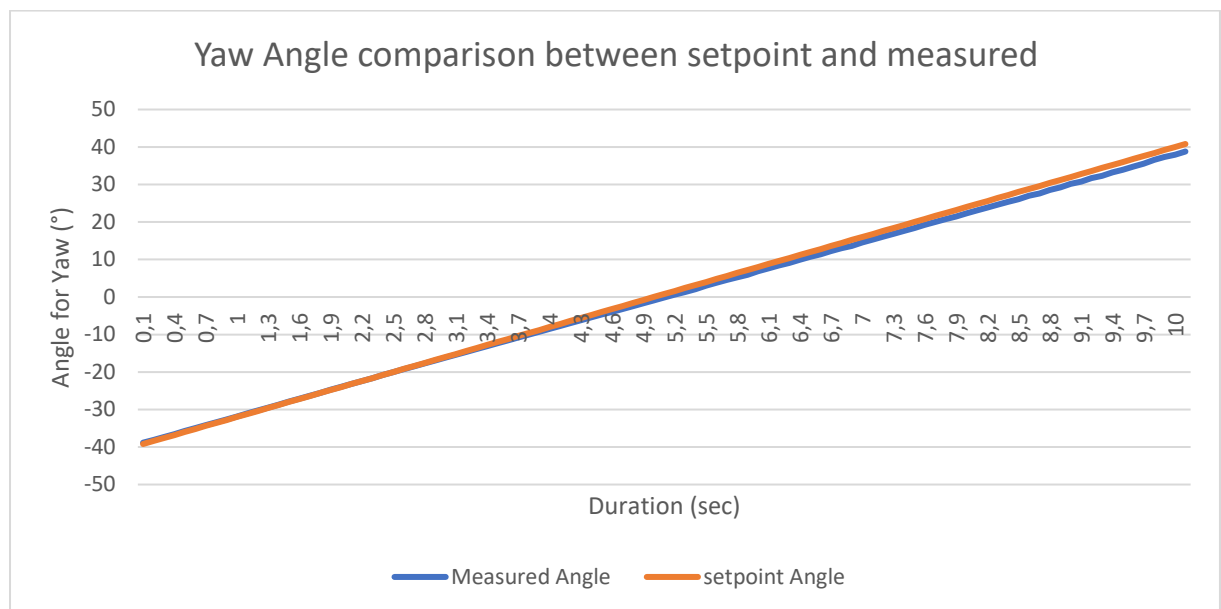


Figure 17 Yaw comparison between yaw setpoint and angle measured.

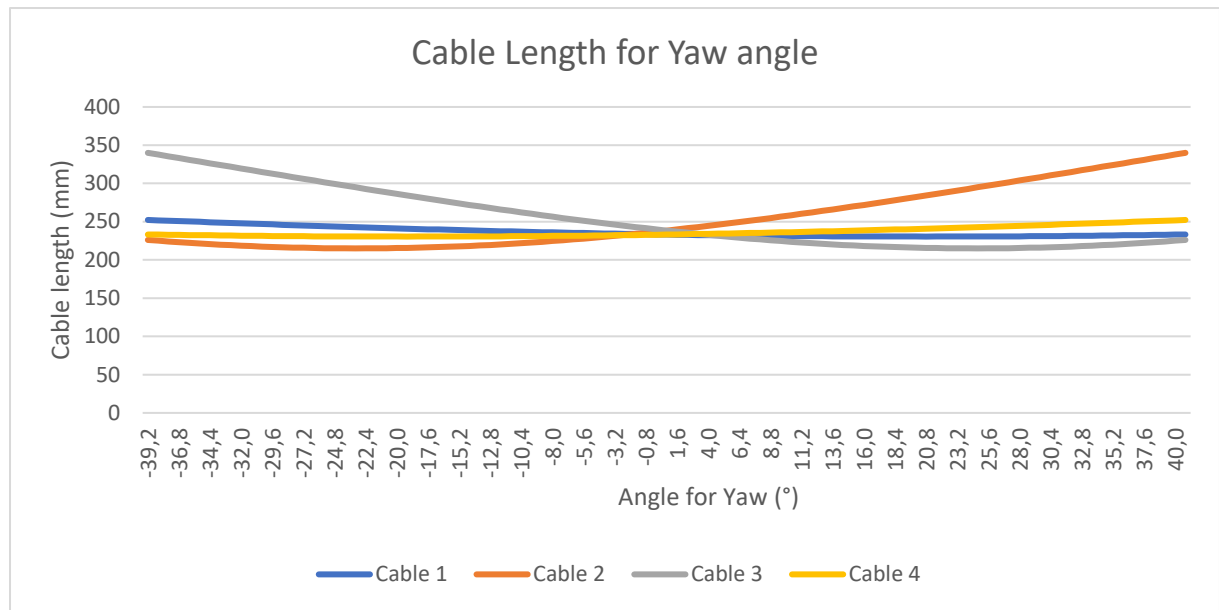


Figure 18 Simulated cable length for yaw angle.

In the angle simulation, a small difference between the setpoint and the measured angle was observed. This variation is mainly because only the gravity force is applied to the foot platform during the simulation excluding the actual weight and dynamic of the foot. The mathematical model calculates the cable lengths for rotations around the ankle, but the simulation is only using these cable lengths to determine the foot platform's rotation and position. Given the structure configuration, same cable lengths can result in multiple platform positions and rotations, considering that the cables remain in tension and that all the joints are spherical. Thus, the position and rotation of the foot platform may not fit the mathematical model's analysis.

4. Prototype and experimental Results

A prototype of the proposed design has been built, Fig. 19. The structure was made with aluminum bars; the foot platform was 3D printed, and the load cells screwed to the cables. To gather the data from the load cells, an Arduino board was used with Analog to Digital Converter for load cell HX711, Fig. 20.

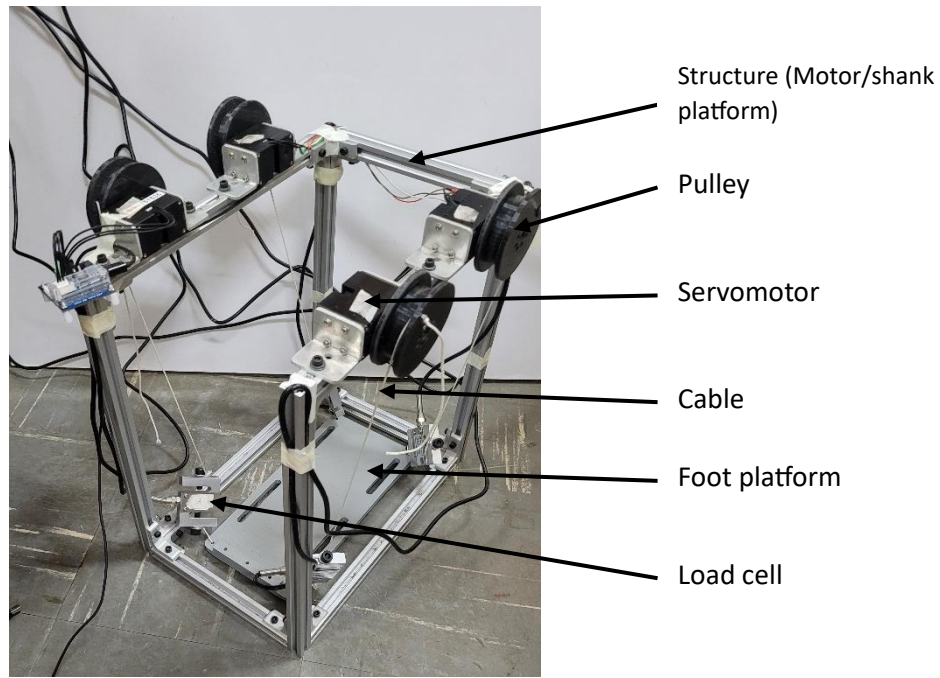


Figure 19 General view of the prototype.

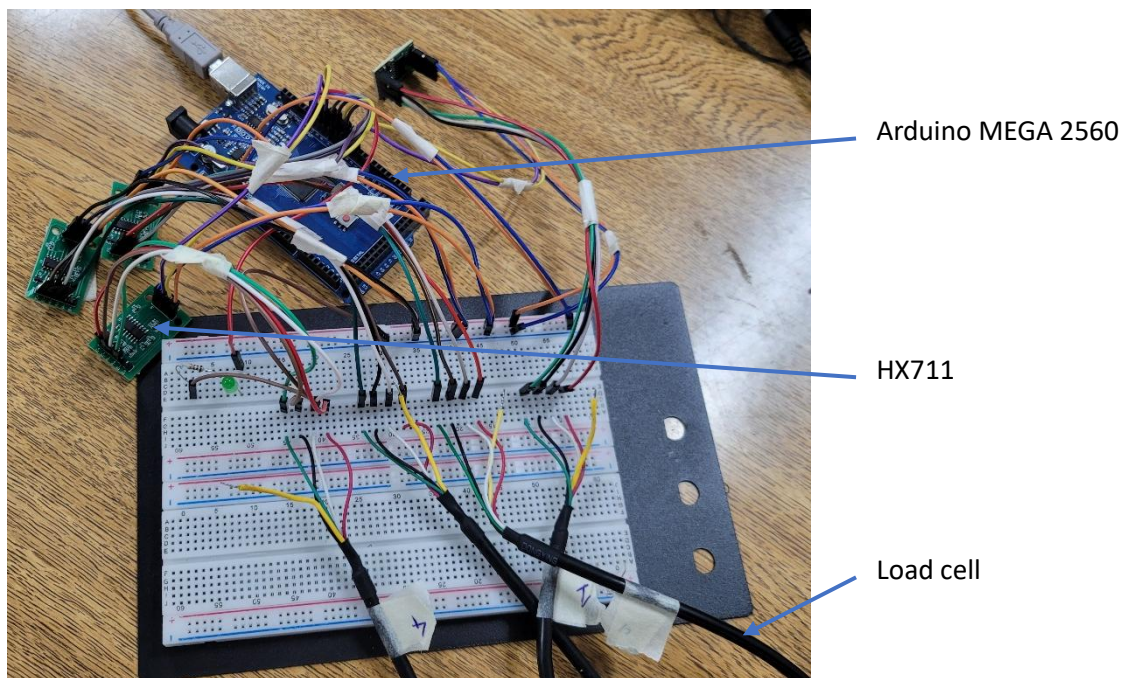


Figure 20 General view of the data gathering unit.

4.1. Control System

The control system of the cable-driven ankle rehabilitation device is composed of two main parts: send the position to the servomotors and get the forces applied in each cable. As the Dynamixel XM540-W270-T are smart servo actuators, the control law is already included inside the motors and then only positions are sent to them [27]. The main process unit is one computer with Matlab, Arduino and C# scripts. The motor positions are sent through the U2D2 communication converter connected in USB. The motors are powered in 12V with a switching adapter and the U2D2 power hub. The load values are converted by HX711 ADCs and gathered with an Arduino MEGA 2560 board connected with

the computer in USB, Fig 21. The Figure 22 represents the conceptual design diagrams of the control system.

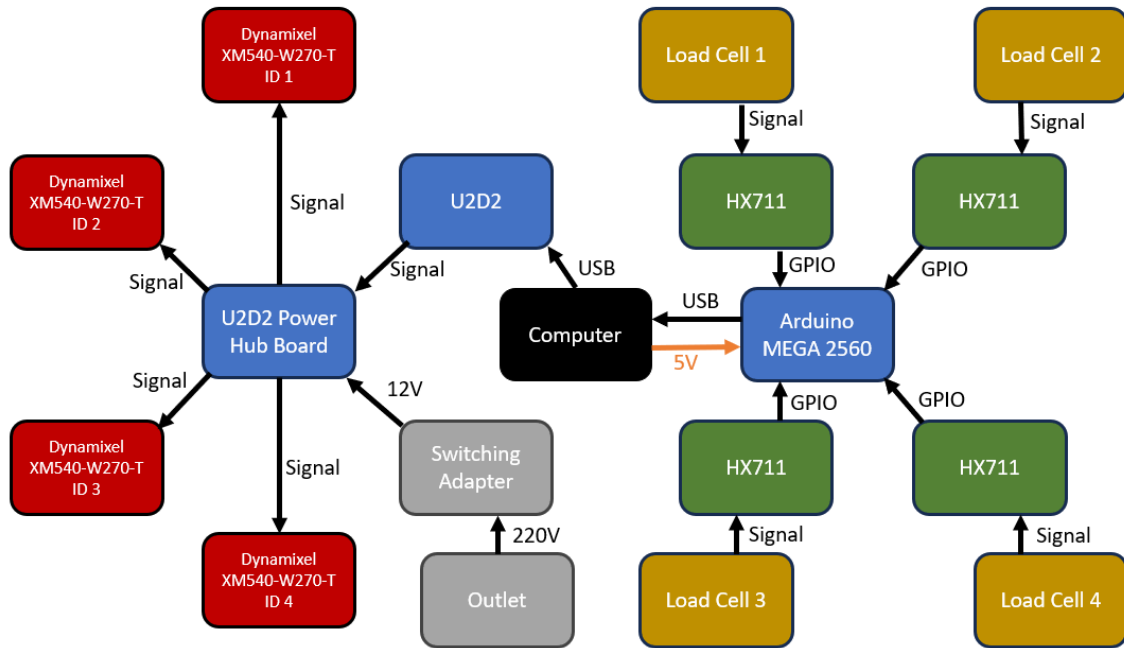


Figure 21 Conceptual design diagram of the data exchange.

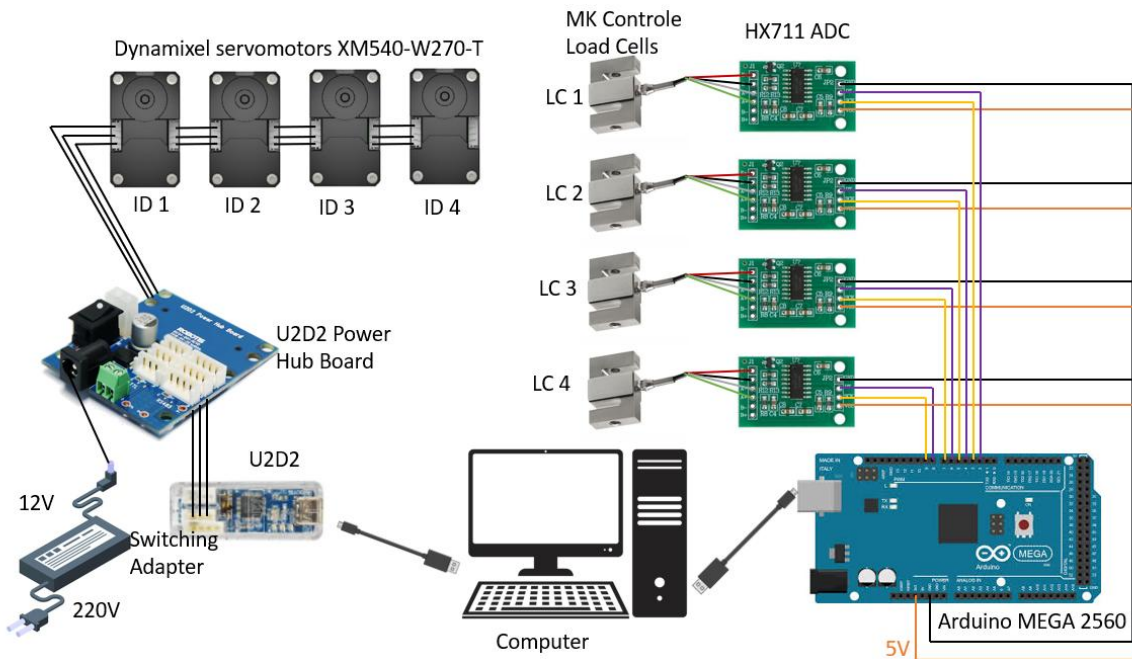


Figure 22 Conceptual design diagram of the connection between the elements.

4.2. Experimental results

Tests were conducted using the prototype. The data were obtained with a digital inclinometer for pitch and roll angles, a cellphone inclinometer for yaw and a graduated rule to measure the height, Fig. 23. The data were gathered on an excel table and plotted in excel charts.

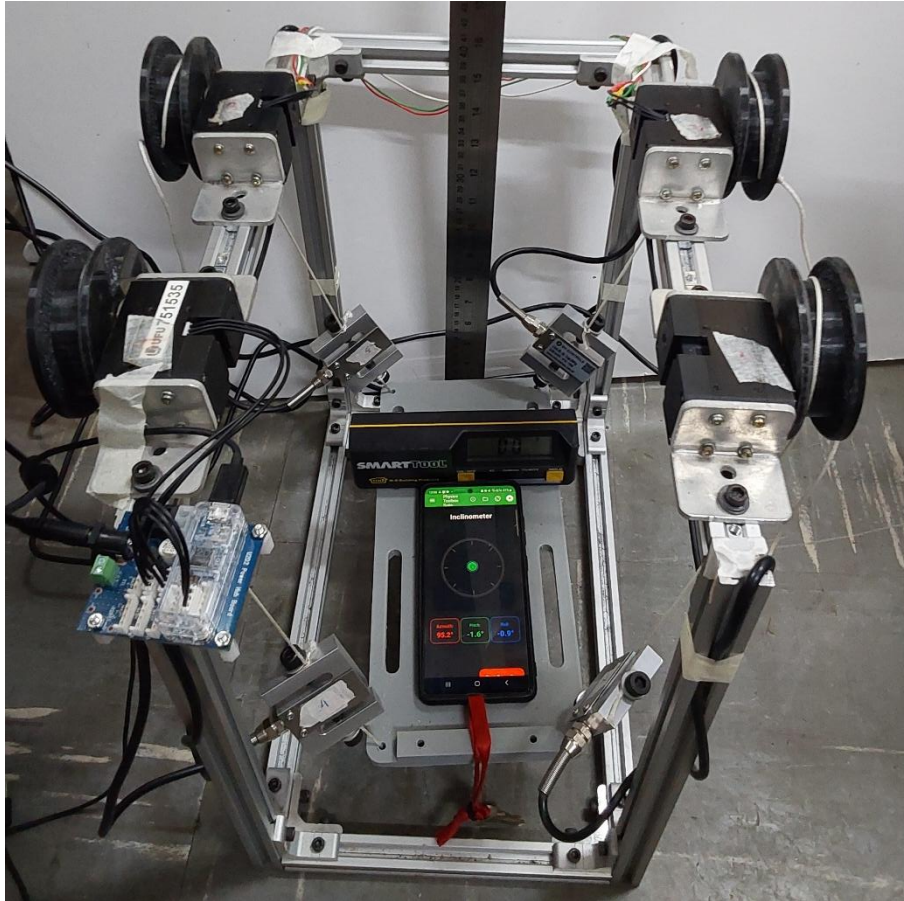


Figure 23 Data gathering protocol.

The first tests were conducted for the height varying between 0 and 225mm and the three angles are set to 0° , Figs. 24 and 25.

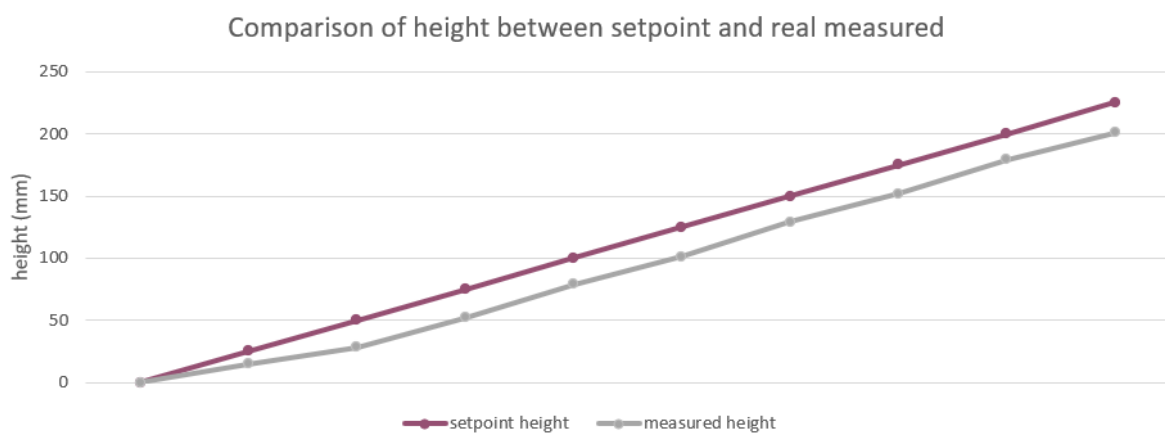


Figure 24 Experimental height comparison between height setpoint and height measured.

The difference between the setpoint and the measured height is due to the elastic properties of the cable used. Indeed, from a set point height of 50mm, the difference stays almost the same, the cables having reached their maximum elasticity. Tests were then conducted for the angles: Fig. 26 for pitch, Fig. 27 for roll and Fig. 28 for yaw. For the three experiences, only one angle is varying, and the others are set to 0° while the height is set to 150mm.

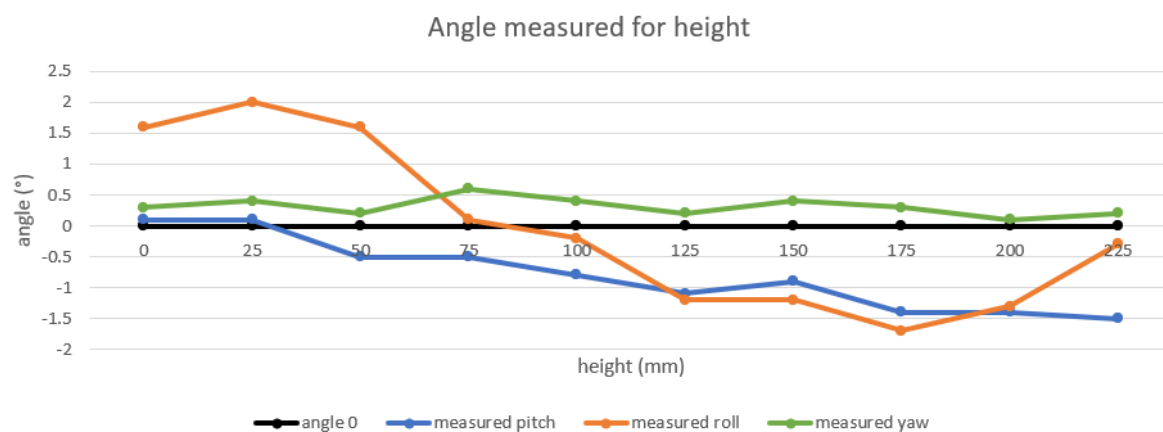


Figure 25 Angle measured for height variation and null target angles.

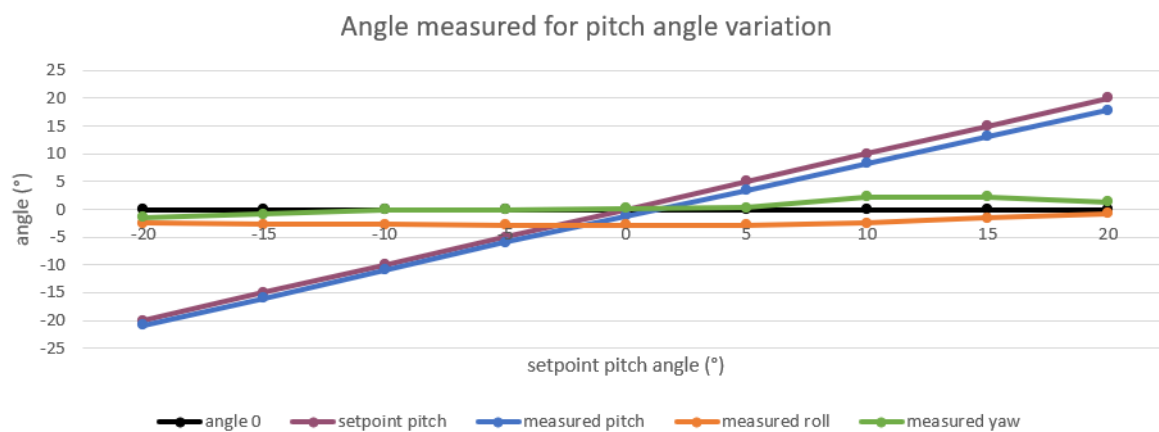


Figure 26 Pitch comparison between pitch setpoint and angle measured.

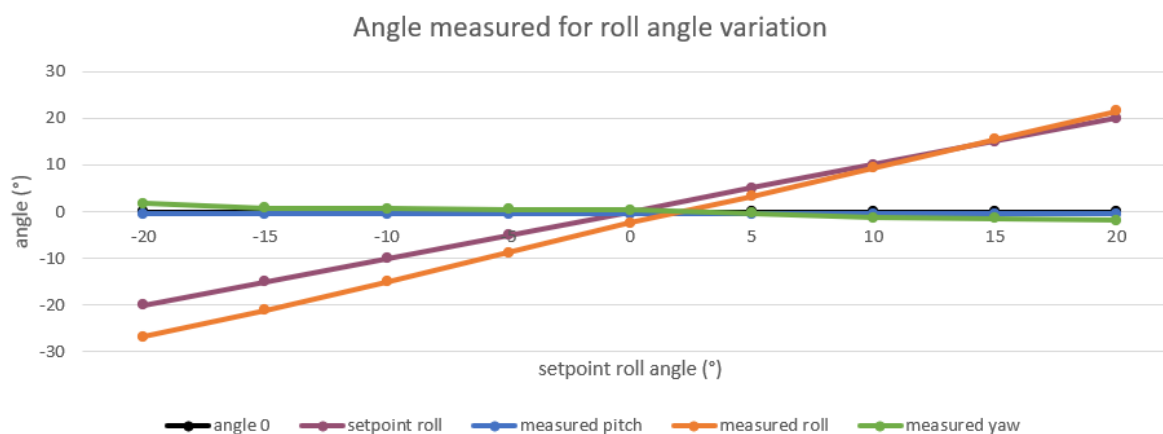


Figure 27 Roll comparison between roll setpoint and angle measured.

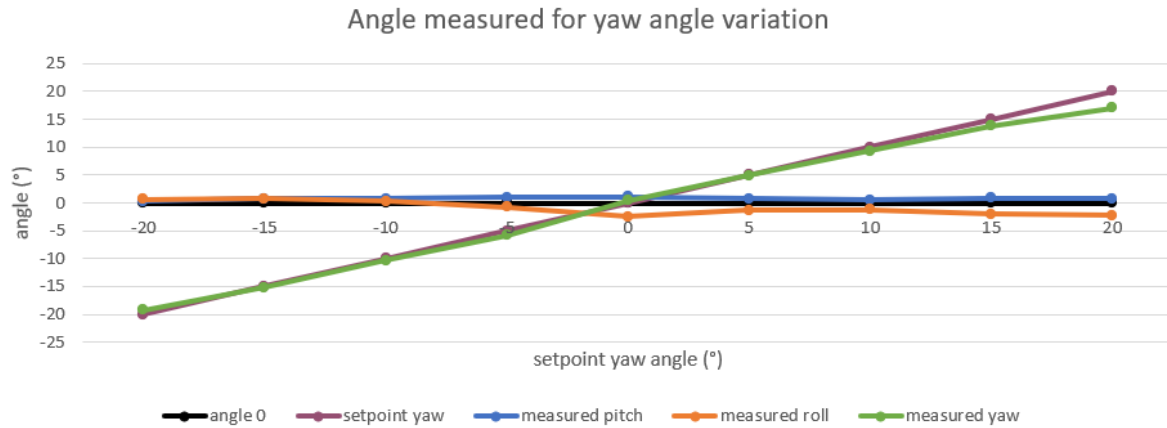


Figure 28 Yaw comparison between yaw setpoint and angle measured.

Like in the Unity's simulations, a small difference between the setpoint and the measured angles can be noticed. These differences are, however, not prejudicial and the tests confirm the viability of the mathematical model used. The CAD/CAE results together with the experimental results demonstrate that the novel proposed cable-driven robot design can operate in the full range of motion of the ankle and with limited variation of the cables lengths without reaching joint limits.

4.3. Serious Game

For the ankle rehabilitation to be more interesting and entertaining, the cable-driven ankle rehabilitation device was combined with a serious game made with Unity 3D. The use of serious games can record data of the rehabilitation section to help the specialists in patient's support. The first serious game made is used for the pitch ankle rehabilitation movement. In this serious game, marble falling must be directed to the correct box using rotative platforms, Fig. 29. The game platforms are connected to the foot platform of the device, so the pitch angle of the foot platform is the same as the rotation of the game platforms.

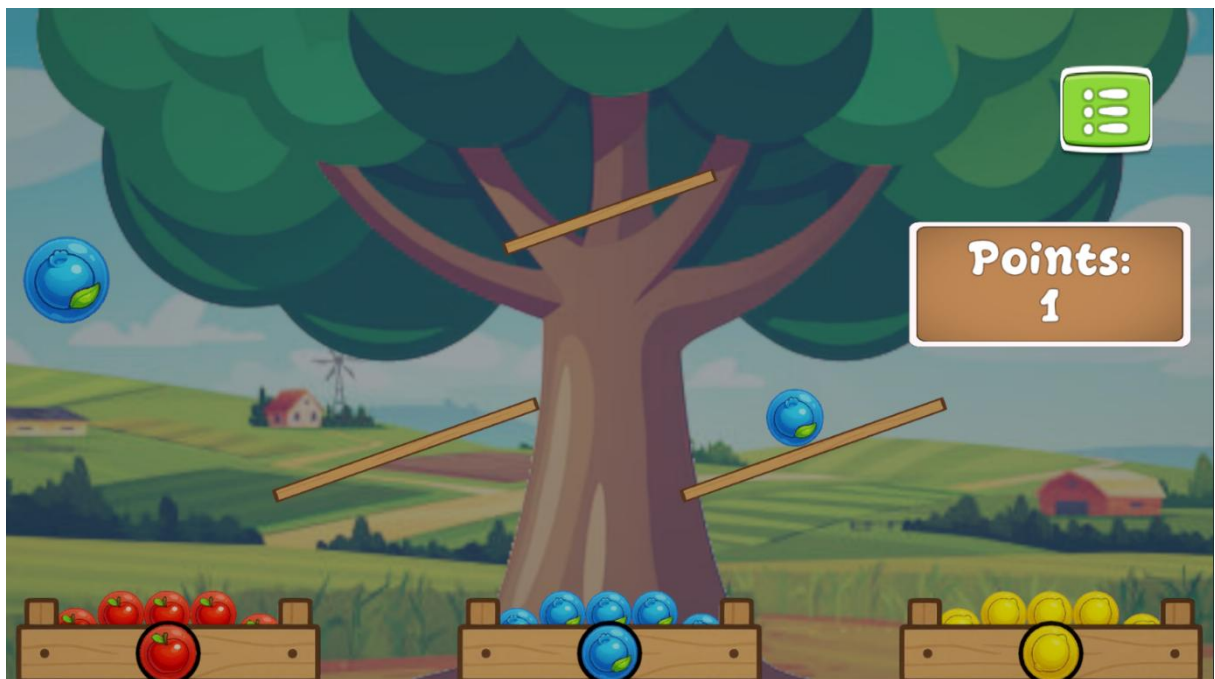


Figure 29 General view of the Marble Balance serious game for pitch ankle rehabilitation.

This serious game has three main game modes: an active mode where the user must move his ankle to play, the passive mode where the game plays on its own and moves the foot platform, and the adaptative mode who is the active mode that helps the user if he has difficulties playing.

The passive mode is controlled by a control law and target angle according to the color of the marble. If the marble is red, the platform will get a target pitch angle of 20° and will reach it with a proportional controller. For the yellow marble, the target angle is -20° and for the blue one, the target angle depends on the rotation of the platform and the position of the marble: if the platform rotation was positive before, the target angle is first 10° then -20° when the marble leaves the higher platform and the opposite, respectively -10° and 20° , if the platform rotation was negative, Fig. 30. The angle of the game platforms is transmitted to the foot platform of the device for passive rehabilitation.

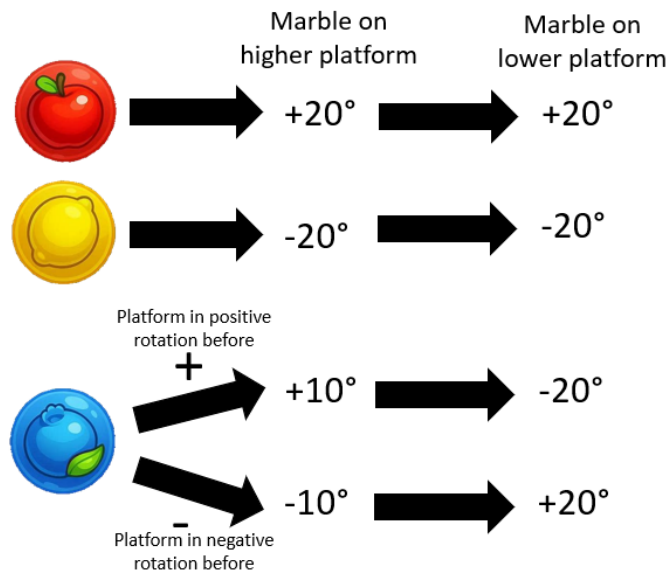


Figure 30 Passive mode.

The active mode uses the load cells to get the ankle movement of the user. For this, the force FS on each cable is first recorded in a stable position with the user's foot and will be used as a reference. Then, by performing tests on the force put on each cable for the different ankle motion, a ratio R has been recorded. The game then uses a linear equation to get the instruction: the force F put on each cable according to FS and R permit to determine A , the user's ankle angle and rotate the platform consequently, **Equation 10**. The sensitivity S can be changed in the settings menu to change the scaling used for the rotation speed, **Equation 11**.

$$A = \frac{F - FS}{R} \quad (10)$$

$$R = R * (0.5 + S); \quad S \in [0; 1] \quad (11)$$

A video is available with the cable-driven ankle rehabilitation device in use combined with the serious game: <https://youtu.be/ODjrXlw1uW8>.

5. Conclusions

Ankle injuries can negatively impact mobility. In this research, a novel variation of the CABLEAnkle device has been analyzed. The proposed design was concept to be lightweight wearable device, easy to set up and adapt to different patients. The device permits the adjustment of the height to be more comfortable with the rehabilitation protocols/exercises.

Despite significant progress, the development of ankle rehabilitation devices still faces important challenges, such as achieving precise alignment with the anatomical ankle axis, providing natural multi-DOF motion in compact and affordable systems, and enabling adaptive, personalized therapy. Many existing platforms also lack robust safety features, ergonomic comfort, and seamless integration into clinical routines. In this context, the cable-driven rehabilitation device presented in this paper offers several advantages that directly address these limitations. Cable transmissions naturally reduce weight and inertia at the end-effector, improving user comfort and safety while facilitating smoother, more compliant interaction. Their flexible routing enables better alignment with the patient's ankle axis, and the proposed multi-cable configurations allow complex motion generation without bulky actuators placed near the joint. Additionally, the lower mechanical complexity and reduced material cost of cable-driven proposed solution contribute to more accessible and scalable design. These characteristics position the proposed cable-driven prototype as a promising approach to overcoming long-standing barriers in ankle rehabilitation technology.

The mathematical model and the simulation/experimental results demonstrated the design's accuracy and expected motion range of the ankle. After simulated results, practical tests were conducted using a prototype, which validated the concept under real-world conditions. A serious game for plantarflexion-dorsiflexion ankle motions was made for the rehabilitation to be more entertaining. The next steps in the work involve developing new serious games and testing them with patients.

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